Semaphores in Producer Consumer Problem: Analysis

- 3 semaphores are used
  - *full* (initially 0) for counting occupied slots
  - *Empty* (initially *N*) for counting empty slots
  - *mutex* (initially 1) to make sure that Producer and Consumer do not access the buffer at the same time.
- Here 2 uses of semaphores
  - Mutual exclusion (mutex)

Block on:	Unblock on:
Producer: insert in full buffer	Consumer: item inserted
Consumer: remove from empty buffer	Producer: item removed

Semaphores: Usage

1. Mutual exclusion
2. Controlling access to limited resource
3. Synchronization

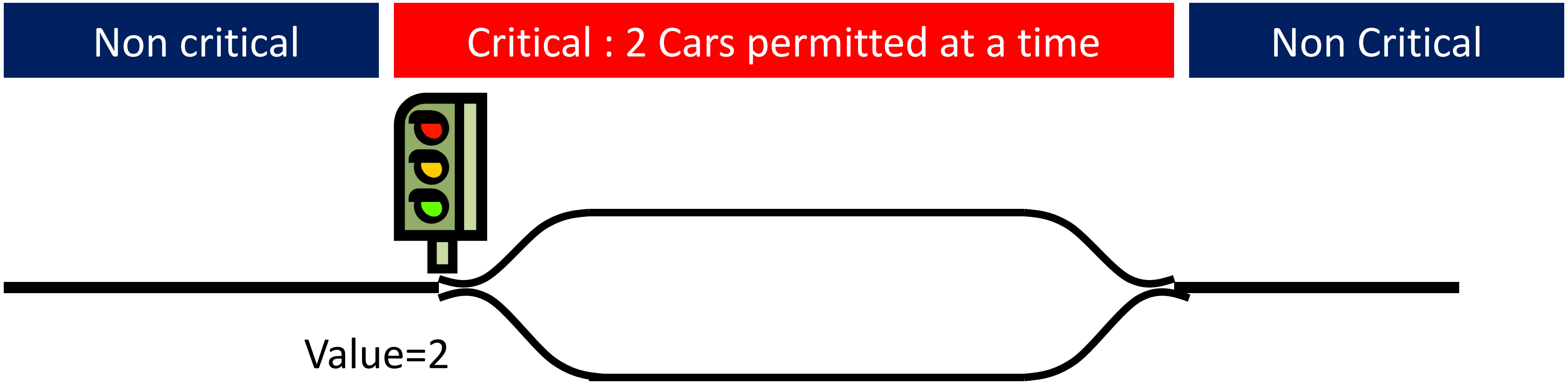
Mutual exclusion

- How to ensure that only one process can enter its C.R.?
- Binary semaphore initialized to 1
- Shared by all collaborating processes
- If each process does a *down* just before entering CR and an *up* just after leaving then mutual exclusion is guaranteed

```
do {
    down(mutex);
    // critical section
    up(mutex);
    // remainder section
} while (TRUE);
```

Controlling access to a resource

- What if we want maximum **m** process/thread can use a resource simultaneously ?
- Counting semaphore initialized to the number of available resources
- Semaphore from railway analogy
  - Here is a semaphore **initialized to 2** for resource control:



Synchronization

- How to resolve dependency among processes
- Binary semaphore initialized to 0
- consider 2 concurrently running processes:
  - P1 with a statement S1 and
  - P2 with a statement S2.
  - Suppose we require that S2 be executed only after S1 has completed.

P1	P2
S1; up(synch);	down(synch); S2;

Semaphores: “Be Careful”

- Suppose the following is done in Producer’s code

```
...
down(&empty)
down(&mutex)
...
```

```
...
down(&mutex)
down(&empty)
...
```

Just the order is reversed

- If buffer full P would block due to down(&empty) with mutex = 0.
- So now if C tries to access the buffer, it would block too due to its down(&mutex).
- Both processes would stay blocked forever: DEADLOCK

Monitors

- A higher level synchronization primitive
- A collection of procedures, variables and data structures grouped in a special kind of module or package.

```
monitor example
        integer i;
        condition c;

        procedure producer( );
        .
        .
        .
        end;

        procedure consumer( );
        .
        .
        .
        end;
end monitor;
```

Example of a monitor

Monitors

- Only one process can be active in a monitor at any instant
- Monitors are programming language construct, so the compiler knows they are special and can handle calls to monitor procedures differently from other calls.
- Because the compiler, not the programmer, is arranging the mutual exclusion, it is **safer**
- We also need a way to block and wakeup: Wait and Signal (done on a condition variables)

Monitors

- *wait* is called on some condition variables:
  - Calling process is blocked
  - another process that had been previously prohibited from entering the monitor is allowed to enter now.
- *signal* is called on some condition variable:
  - A process waiting on *that CV* is given the chance to get up.
  - Who should run? Caller or awakened one?

Alternative#1: Let newly awakened process to run suspending the caller.

Alternative#2: Process doing a signal must exit the monitor immediately i.e. signal statement may appear only as the final statement in a monitor procedure

Outline of producer-consumer using Monitors

```
monitor ProducerConsumer
    condition full, empty;
    integer count;
    procedure insert(item: integer);
    begin
        if count = N then wait(full);
        insert_item(item);
        count := count + 1;
        if count = 1 then signal(empty)
    end;
    function remove: integer;
    begin
        if count = 0 then wait(empty);
        remove = remove_item;
        count := count − 1;
        if count = N − 1 then signal(full)
    end;
    count := 0;
end monitor;
```

```
procedure producer;
begin
    while true do
    begin
        item = produce_item;
        ProducerConsumer.insert(item)
    end
end;
procedure consumer;
begin
    while true do
    begin
        item = ProducerConsumer.remove;
        consume_item(item)
    end
end;
```

- Outline of producer-consumer problem with monitors
  - only one monitor procedure active at one time
  - buffer has N slots

Producer-consumer solution in Java

```
public class ProducerConsumer {
    static final int N = 100;      // constant giving the buffer size
    static producer p = new producer( );    // instantiate a new producer thread
    static consumer c = new consumer( ); // instantiate a new consumer thread
    static our_monitor mon = new our_monitor( );     // instantiate a new monitor

    public static void main(String args[ ]) {
      p.start( );      // start the producer thread
      c.start( );      // start the consumer thread
    }

    static class producer extends Thread {
      public void run( ) {// run method contains the thread code
        int item;
        while (true) {      // producer loop
          item = produce_item( );
          mon.insert(item);
        }
      }
      private int produce_item( ) { ... }      // actually produce
    }

    static class consumer extends Thread {
      public void run( ) { run method contains the thread code
        int item;
        while (true) {      // consumer loop
          item = mon.remove( );
          consume_item (item);
        }
      }
      private void consume_item(int item) { ... }// actually consume
    }
```

```
    static class our_monitor {   // this is a monitor
       private int buffer[ ] = new int[N];
       private int count = 0, lo = 0, hi = 0;   // counters and indices

       public synchronized void insert(int val) {
          if (count == N) go_to_sleep( );     // if the buffer is full, go to sleep
          buffer [hi] = val; // insert an item into the buffer
          hi = (hi + 1) % N;        // slot to place next item in
          count = count + 1;        // one more item in the buffer now
          if (count == 1) notify( );        // if consumer was sleeping, wake it up
       }

       public synchronized int remove( ) {
          int val;
          if (count == 0) go_to_sleep( );     // if the buffer is empty, go to sleep
          val = buffer [lo]; // fetch an item from the buffer
          lo = (lo + 1) % N;        // slot to fetch next item from
          count = count - 1;        // one few items in the buffer
          if (count == N - 1) notify( ); // if producer was sleeping, wake it up
          return val;
      }
      private void go_to_sleep( ) { try{wait( );} catch(InterruptedException exc) {};}
    }
}
```

Problems with monitors and semaphores

- Semaphores are too low level
- Monitors are not usable except in a few programming languages
- Designed to work in an environment having access to a common memory
- Doesn't allow information exchange among machines
- None of them would work in a distributed systems (why?) consisted of multiple CPUs, each with its own private memory connected by a LAN.

Message Passing

- solution to the problem of semaphores and monitors w.r.t distributed systems
- A method of IPC that uses two primitives
  - send and receive: system calls.
  - send(destination, &message)
  - receive(source, &message)
  - If no message is available:
    - The receiver can block until one arrives
    - Return immediately with an error code

Message Passing

- Challenges (Study of Computer Networks)
  - Messages can be lost by the network.
  - Acknowledgement and retransmission issue.
  - Process naming.
  - Authentication.
  - Performance issue.

Producer Consumer with Message Passing

Assumptions:

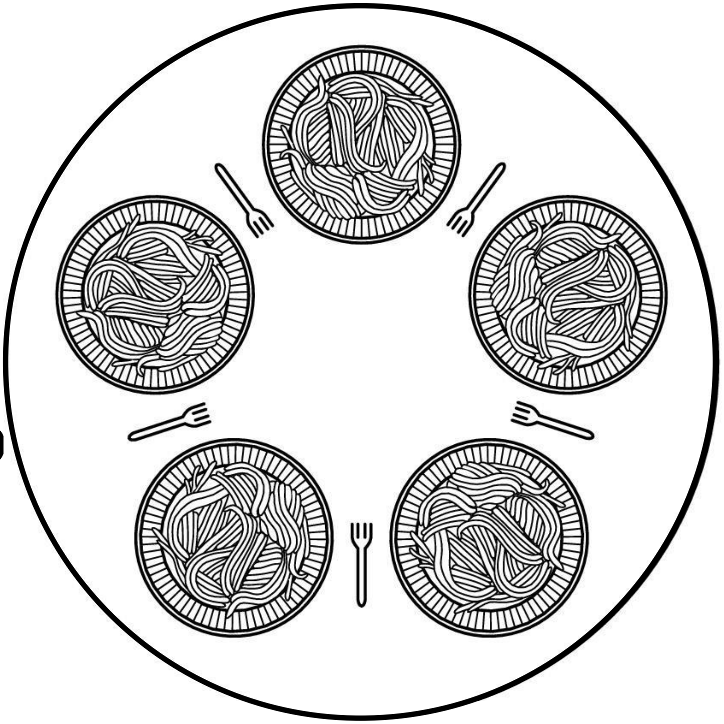
- All messages are of same size
- Messages sent but not yet received are automatically buffered

```
#define N 100                                  /* number of slots in the buffer */

void producer(void)
{
        int item;
        message m;                             /* message buffer */

        while (TRUE) {
              item = produce_item( );          /* generate something to put in buffer */
              receive(consumer, &m);           /* wait for an empty to arrive */
              build_message(&m, item);         /* construct a message to send */
              send(consumer, &m);              /* send item to consumer */
        }
}

void consumer(void)
{
        int item, i;
        message m;

        for (i = 0; i < N; i++) send(producer, &m);  /* send N empties */
        while (TRUE) {
              receive(producer, &m);           /* get message containing item */
              item = extract_item(&m);         /* extract item from message */
              send(producer, &m);              /* send back empty reply */
              consume_item(item);              /* do something with the item */
        }
}
```

The producer-consumer problem with N messages

Dining Philosophers

An example problem for process synchronization



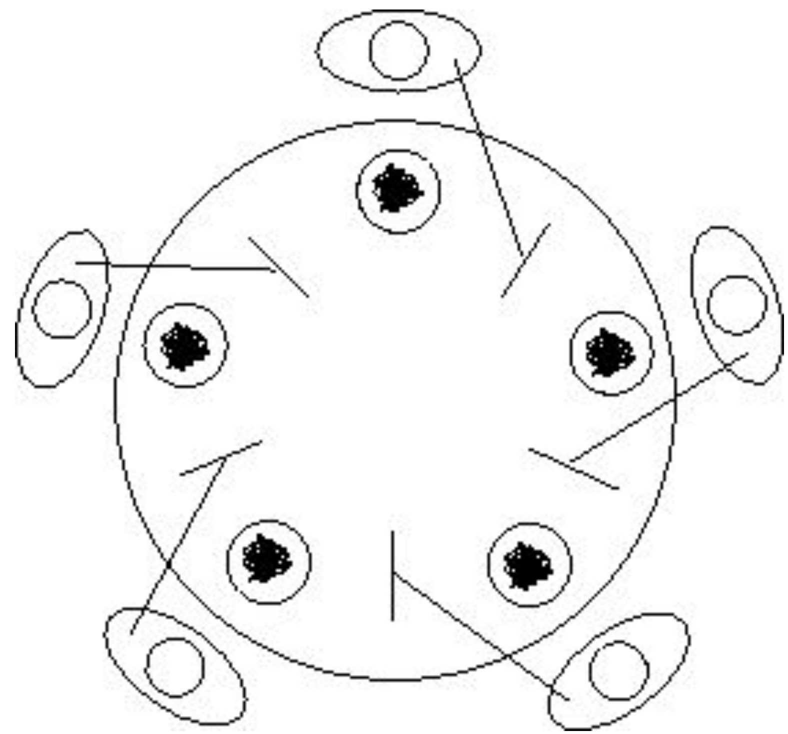
- Philosophers spend their lives thinking and eating
- Don't interact with their neighbors
- When get hungry try to pick up 2 chopsticks (one at a time in either order) to eat
- Need both to eat, then release both when done
- How to program the scenario avoiding all concurrency problems?

Dining Philosophers: A Solution

```
#define N 5                          /* number of philosophers */

void philosopher(int i)              /* i: philosopher number, from 0 to 4 */
{
    while (TRUE) {
        think( );                    /* philosopher is thinking */
        take_fork(i);                /* take left fork */
        take_fork((i+1) % N);        /* take right fork; % is modulo operator */
        eat( );                      /* yum-yum, spaghetti */
        put_fork(i);                 /* put left fork back on the table */
        put_fork((i+1) % N);         /* put right fork back on the table */
    }
}
```

Dining Philosophers: Problems with Previous Solution

- Deadlock may happen
- Does this solution prevents any such thing from happening ?
  - Everyone takes the left fork simultaneously



Dining Philosophers: Problems with Previous Solution

Tentative Solution:

- After taking left fork, check whether right fork is available.
- If not, then return left one, wait for some time and repeat again.

Problem:

- All of them start and do the algorithm synchronously and simultaneously:
- STARVATION (A situation in which all the programs run indefinitely but fail to make any progress)
- Solution: Random wait; but what if the most unlikely of same random number happens?

Another Attempt, Successful!

```
void philosopher(int i)
{
  while (true)
  {
      think();
      down(&mutex);
      take_fork(i);
      take_fork((i+1)%N);
      eat();
      put_fork(i);
      put_fork((i+1)%N);
      up(&mutex);
  }
}
```

- Theoretically solution is OK- no deadlock, no starvation.
- Practically with a performance bug:
  - Only one philosopher can be eating at any instant: absence of parallelism

Final Solution part 1

```
#define N           5              /* number of philosophers */
#define LEFT        (i+N−1)%N      /* number of i's left neighbor */
#define RIGHT       (i+1)%N        /* number of i's right neighbor */
#define THINKING    0              /* philosopher is thinking */
#define HUNGRY      1              /* philosopher is trying to get forks */
#define EATING      2              /* philosopher is eating */
typedef int semaphore;             /* semaphores are a special kind of int */
int state[N];                      /* array to keep track of everyone's state */
semaphore mutex = 1;               /* mutual exclusion for critical regions */
semaphore s[N];                    /* one semaphore per philosopher */

void philosopher(int i)            /* i: philosopher number, from 0 to N−1 */
{
    while (TRUE) {                 /* repeat forever */
        think( );                  /* philosopher is thinking */
        take_forks(i);             /* acquire two forks or block */
        eat( );                    /* yum-yum, spaghetti */
        put_forks(i);              /* put both forks back on table */
    }
}
```

Final Solution Part 2

```
void take_forks(int i)              /* i: philosopher number, from 0 to N−1 */
{
    down(&mutex);                   /* enter critical region */
    state[i] = HUNGRY;              /* record fact that philosopher i is hungry */
    test(i);                        /* try to acquire 2 forks */
    up(&mutex);                     /* exit critical region */
    down(&s[i]);                    /* block if forks were not acquired */
}

void put_forks(i)                   /* i: philosopher number, from 0 to N−1 */
{
    down(&mutex);                   /* enter critical region */
    state[i] = THINKING;            /* philosopher has finished eating */
    test(LEFT);                     /* see if left neighbor can now eat */
    test(RIGHT);                    /* see if right neighbor can now eat */
    up(&mutex);                     /* exit critical region */
}

void test(i)                        /* i: philosopher number, from 0 to N−1 */
{
    if (state[i] == HUNGRY && state[LEFT] != EATING && state[RIGHT] != EATING) {
        state[i] = EATING;
        up(&s[i]);
    }
}
```

The Readers and Writers Problem

- Dining Philosopher Problem: Models processes that are competing for exclusive access to a limited resource
- Readers Writers Problem: Models access to a database

Example: An airline reservation system- many competing process wishing to read and write-

  - Multiple readers simultaneously- acceptable
  - Multiple writers simultaneously- not acceptable
  - Reading, while write is writing- not acceptable

The Readers and Writers Problem

- *First* variation – no reader kept waiting unless writer has permission to use shared object
- *Second* variation – once writer is ready, it performs write ASAP

```c
typedef int semaphore;                  /* use your imagination */
semaphore mutex = 1;                    /* controls access to 'rc' */
semaphore db = 1;                       /* controls access to the database */
int rc = 0;                             /* # of processes reading or wanting to */

void reader(void)
{
     while (TRUE) {                     /* repeat forever */
          down(&mutex);                 /* get exclusive access to 'rc' */
          rc = rc + 1;                  /* one reader more now */
          if (rc == 1) down(&db);       /* if this is the first reader ... */
          up(&mutex);                   /* release exclusive access to 'rc' */
          read_data_base( );            /* access the data */
          down(&mutex);                 /* get exclusive access to 'rc' */
          rc = rc – 1;                  /* one reader fewer now */
          if (rc == 0) up(&db);         /* if this is the last reader ... */
          up(&mutex);                   /* release exclusive access to 'rc' */
          use_data_read( );             /* noncritical region */
     }
}


void writer(void)
{
     while (TRUE) {                     /* repeat forever */
          think_up_data( );             /* noncritical region */
          down(&db);                    /* get exclusive access */
          write_data_base( );           /* update the data */
          up(&db);                      /* release exclusive access */
     }
}
```

Issue regarding the solution

- Inherent priority to the readers
- Say a new reader arrives every 2 seconds and each reader takes 5 seconds to do its work. What will happen to a writer?

Issue regarding second variation

- Writer don’t have to wait for readers that came along after it
- Less concurrency, lower performance